Wireless Personal Communications
https://doi.org/10.1007/s11277-021-08994-z

Machine Learning Approaches for Anomaly Detection in IoT: An Overview and Future Research Directions

Nusaybah Alghanmi[1] · **Reem Alotaibi**[1] · **Seyed M. Buhari**[1]

Accepted: 8 August 2021
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

The internet of things (IoT) is the networking of interrelated devices and sensors connected through the internet to transfer and share data. The data gathered from these devices may have anomalies or other errors for various reasons, such as malicious activities or sensor failures. Anomaly detection is found in several domains, such as fault detection and health monitoring systems. In this paper, we review and analyze the relevant literature on existing anomaly detection techniques that apply different machine learning approaches in the IoT. In addition, we examine different anomaly detection datasets used in IoT and highlight the most concerning issues with these datasets for different approaches, and list several future research directions. We believe this survey will serve as a starting point for researchers to gain knowledge from the IoT that employs machine learning approaches to detect anomalies. Moreover, the datasets that associate with anomaly detection in IoT, issues and future directions from a dataset perspective.

Keywords Internet of things (IoT) · Machine learning algorithms · Anomaly detection · Sensor data

Abbreviations

IoT	Internet of things
IWC	Inverse weight clustering
DBSCAN	Density-based spatial clustering of applications with noise
RF	Random forest
LDA	Linear discriminant analysis
PCA	Principal component analysis
NB	Naive bayes
FF	Farthest first

✉ Nusaybah Alghanmi
nalghanmi0020@stu.kau.edu.sa

Reem Alotaibi
ralotibi@kau.edu.sa

Seyed M. Buhari
mesbukary@kau.edu.sa

[1] Department of Information Technology, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia

Published online: 27 August 2021

EM	Expectation maximization
SVDD	Support vector data description
DNNs	Deep neural networks
DT	Decision tree
DOS	Denial of service
MD	Mahalanobis distance
LOF	Local outlier factor
KNNs	K-nearest neighbours
SVMs	Support vector machines
NNs	Neural networks
OC-SVM	One-class SVM
RBF	Radial basis function
FPR	False positive rate
TPR	True positive rate

1 Introduction

The internet of things (IoT) is an emerging environment that consists of interrelated devices and sensors connected through the Internet. It has gained attention in different sectors, for instance academia and business, due to the advantages of enabling 'smart living'. The IoT can be applied in different application domains, such as smart homes, wearable devices, smart cities, healthcare, agriculture, transportation, and industry.

For example, in smart cities, streetlights can be controlled by collecting data from a sensor embedded in the light (things or objects) then sending this information via the Internet to the head station to decide the level of the required lighting, based on time of the day, to save energy. The main advantage of the IoT is that it helps to make appropriate decisions based on the data collected by sensors, and tracks 'things' in a smart way.

Cisco has estimated that IoT-generated data will amount to 847 Zettabytes (ZB) by 2021, though the amount of data stored will be appreciably less at roughly 7.2 ZB [1]. The IoT generates large volumes of data which, for numerous reasons, may include abnormal patterns or anomalous behaviours including malevolent activities or failures of sensors.

In this context, anomalous refers to an abnormality in data which is different to a predicted pattern [2]. Characteristics which may be classed as anomalous differ from the norm and will rarely be found in the data [3]. Anomaly detection is a technique to discover unusual behaviours which are distinct from the predicted patterns. It is frequently deployed in multiple applications, including intrusion detection, error and fraud detection, and systems used for monitoring an individual's health.

Machine learning techniques are one form of these techniques to perform anomaly detection in the IoT. In this approach, the learning tasks are executed in either a supervised or unsupervised mode. Supervised learning employs known data (labelled training data) which enables one to classify data into normal and abnormal classes, while unsupervised learning handles unlabelled data through identification of similarities within the data, which means that normal and abnormal classes are, as such, unknown.

The incentive driving this work is that there are innumerable devices and sensors which are connected to the IoT, weather sensors being a classic example, and which are responsible for the generation of considerable amounts of data for collection and analysis. This data is the foundation stone of any system which is designed in order to

make intelligent decisions on plans and actions in the future. A principal challenge in any anomaly detection system is that there is not always ready availability of labelled data [2]. Methods using a supervised mode for anomaly detection need data which has been labelled [2, 4], though class labels (normal or anomalous) for IoT data are rarely available.

This paper presents an overview of the existing solutions that apply machine learning methods to detect anomalies using labelled and unlabelled data. It also analyzes anomaly detection datasets used in the IoT and discusses the issues related to these datasets. In this way, this paper helps the researchers to achieve the following:

- Facilitate a better understanding of how anomaly detection is performed using machine learning techniques.
- Survey and categorize the current anomaly detection solutions using machine learning approaches in the IoT.
- Analyze the IoT datasets for anomaly detection, and list challenges and possible future research directions associated with these datasets.

The paper is organized as follows: Section 2 presents the background to machine learning algorithms for anomaly detection in the IoT. Section 3 reviews the current anomaly detection models used in the IoT, drawing on the literature in this area. Section 4 presents the current anomaly detection datasets and their associated issues. Finally, Sects. 5 and 6 list future research directions and present the conclusion, respectively.

2 Background

Anomaly detection methods seek to detect unusual behaviours that do not conform to predicted behaviours [2], where machine learning algorithms are used to identify the class label or target for the data (normal/anomaly). Models are constructed using different keys, e.g. datasets, learning algorithm type, selection of feature, and techniques used for evaluation.

- The dataset is a collection of data obtained from IoT devices. The data (matrix) consists of observations, samples or instances (in rows), and attributes or features (in columns). In this paper, these are treated as synonymous and are used interchangeably.
- The learning algorithm can be either supervised or unsupervised.
- Feature selection is a method used to select a subset of features or attributes for building the model.
- Evaluation techniques are employed to validate the results of machine learning algorithms.

Figure 1 shows the machine learning workflow used to detect anomalies in the IoT. After it is gathered from IoT devices, the data is stored. Later, machine learning algorithms (supervised or unsupervised) are used for anomaly detection. The performance evaluation is the final step, to validate and evaluate the proposed models.

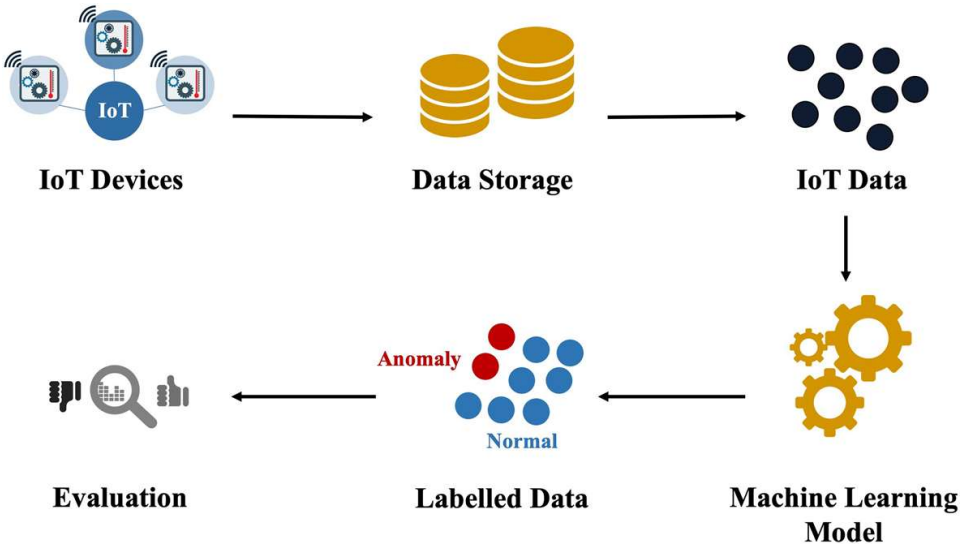


Fig. 1 Machine learning workflow for anomaly detection in IoT

3 Literature Review

The literature was classified into categories based on the scope of the research papers. Systematic and non-systematic approaches were adopted during the literature search stage. At first, some articles were explored. Later on, documents related to anomaly detection in the IoT were found via searches of databases such as Web of Science, Google Scholar, Scopus, and Science Direct. More importantly, most of the reviewed literature was accessed through backwards citations of identified papers.

We applied the following search terms: anomaly detection, machine learning, and sensor data or IoT. The databases used in the search were Google Scholar, IEEE Xplore and Web of Science. The selected studies were mostly published between 2014 and 2021.

This section provides solutions for anomaly detection in IoT that current apply supervised or unsupervised learning algorithms. Figure 2 shows that supervised learning can be separated into single, ensemble, and two-tier models. Sections 3.1.1 and 3.1.2 will explain this in greater detail.

3.1 Supervised Learning

Supervised learning includes two practical steps: training and testing. The former trains the labelled data to construct the model, while the latter predicts the class labels (anomaly/

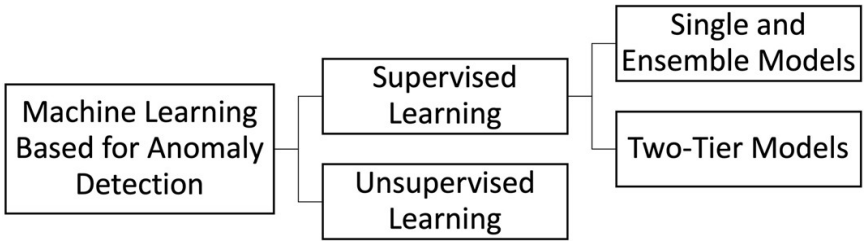


Fig. 2 Machine learning approaches for IoT data

normal) for unknown data. There is a wide range of supervised learning algorithms, such as decision tree (DT), naive bayes (NB), and K-nearest neighbours (KNNs).

Three approaches that could be applied to build a model for the class labels prediction are described by Tsai et al. [5]; these are the single, ensemble or two-tier models. In single models, only one learner is used, such as support vector machines (SVMs). Ensemble models enhance the performance by including a variety of weak learners, whereas, two-tier models consist of different learners, similar to the neuro-fuzzy method. Two-tier models build the model in two steps [6], starting with using the data to initiate intermediate results, and then as input for producing the final outcomes.

3.1.1 Single and Ensemble Models

Single models use only one learner to classify the data. Ford et al. [7] modelled energy consumption behaviour (ECB) using neural networks (NNs). Using data from the Irish Social Science Data Archive (ISSDA) [8], the authors found that NNs performed well, with a 93.75% true positive rate (TPR). The application of NNs was also proposed by Cañedo and Skjellum [9] to secure the IoT network. The authors applied the model to simulate an IoT network consisting of ten sensors and 4,000 observations. NNs were trained twice under different input cases, such as device ID and sensor value, in addition to the delay value in the second case. In both cases, the authors found that the model achieved over 99% prediction accuracy, with $< 1\%$ false negatives in the second case.

To classify real Citypulse project data for Aarhus, Denmark, Jain and Shah [10] deployed multiple learners including NNs, binary SVMs, and multiclass SVMs, [11, 12]. Multiple kernels were deployed for binary and multiclass SVMs, e.g. linear, polynomial, and radial basis function (RBF). In general terms, SVMs case accuracy was higher than that for NNs. To expand, binary SVMs with the RBF outperformed the linear kernel where accuracy was concerned, while multiclass SVMs with a polynomial kernel outperformed alternative kernels.

Hodo et al. [13] proposed offline intrusion detection based on the anomaly detection concept for an IoT network. They used feed-forward NNs with three layers to detect Distributed Denial of Service (DDoS/DoS) attacks. The authors validated the model on a simulated IoT network, and found that the accuracy was 99.4%. Furthermore, Pachauri and Sharma [14] used random forest (RF) as an ensemble model, and KNNs and J48 decision trees as single models. The solution was proposed for detecting anomalies in the medical field by using the MIMIC dataset from the PhysioNet database [15, 16]. The authors found that the RF was better than other selected learners in terms of the receiver operating characteristic (ROC) curve.

Hasan et al. [17] used several machine learning approaches for attack and anomaly detection, applying them to the DS2OS traffic traces dataset [18]. The machine learning approaches included logistic regression (LR), DT, SVMs, RF, and NNs. The results demonstrated that RF, DT, and NNs performed better than other methods, achieving 99.4% accuracy.

3.1.2 Two-Tier Models

The two-tier model classifies the data by combining several learners. Pajouh et al. [19] proposed a two-tier anomaly detection model to detect malicious activities in the IoT backbone. The model, called TDTC, was applied to the NSL-KDD dataset [20], and

consisted of two steps: two-layer dimension reduction and two-tier classification. Linear discriminant analysis (LDA) and principal component analysis (PCA) were employed in two-layer dimension reduction. This step helps to reduce the dimensionality of the data from a complex to a lower dataset. Both NB and KNNs were employed to classify the data in the two-tier classification. At this stage, the data were passed to NB to detect the anomaly behaviours, as shown in Fig. 3 [19]. As NB is a weak learner [21], the normal observations were used to train the certainty-factor version of KNN (CF-KNN) to classify the data into normal and anomaly classes. The results showed that the TDTC achieved the highest accuracy, with 84.86%.

The inverse weight clustering (IWC) and C4.5 algorithms were proposed by Alghuried [22] as a two-tier model for anomaly detection that was applied to Intel Lab data [23]. The instances were grouped into normal and anomaly clusters using IWC, and then passed as labelled data to train the C4.5 algorithm. The results showed that the accuracy was 97%, and the recall was 98.3%. Zhao et al. [24] used PCA to reduce the dimensionality of the dataset and compared two algorithms, KNNs and softmax regression. They applied the proposed model to the KDD Cup 99 dataset [25]. The results showed that the performance of both algorithms after performing PCA was similar in terms of accuracy, false positive rate (FPR), and detection rate, but there was a significant change in the computing complexity, with 5071.31 seconds and 5.72 seconds for KNNs and softmax regression, respectively.

The density-based spatial clustering of applications with noise (DBSCAN) and SVMs was modelled by Emadi and Mazinani [26] on Intel Lab data [23]. After selecting the input features, namely temperature, humidity, and voltage, the proposed model consists of two practical steps. First, the optimal values for the Epsilon and MinPts were configured using the coefficient correlation (CC). In this step, the data was grouped, based on density, into normal and anomaly classes. Second, the normal data gained was then used to train the SVMs algorithm. The accuracy achieved across different experiments was over 94%.

Hosseini and Borojeni [27] proposed a hybrid model based on k-means clustering and sequential minimal optimization (SMO) classification that was applied to Intel Lab data [23]. The data were grouped using k-means clustering into normal and anomaly clusters and then passed to SMO to train it. The results demonstrated that the proposed model achieved 100% accuracy and positive detection rate and 0% FPR.

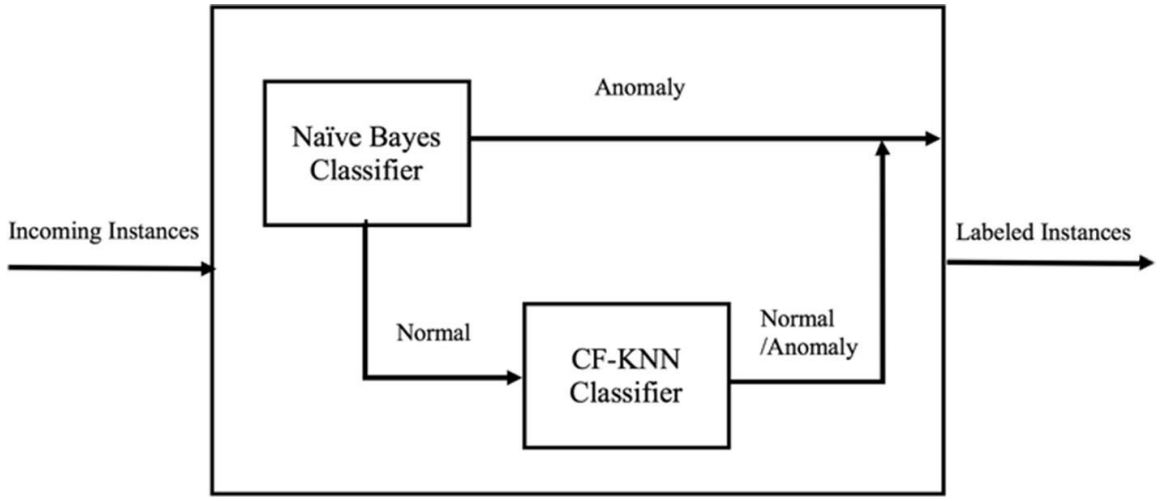


Fig. 3 Classification model in TDTC

Authors in [28] proposed a hybrid learning anomaly detection model for unlabeled data in IoT called HLMCC. The proposed model combines hierarchical affinity propagation (HAP) clustering algorithm followed by DT classification technique. The clustering groups the available data into corresponding clusters (normal and abnormal). After that, this data is used for training the DT algorithm. The model was applied to Labelled Wireless Sensor Network Data Repository (LWSNDR) [29] and landsat satellite dataset [3]. The results demonstrated that the proposed approach enhanced the FPR, recall, precision, and the area under the precision-recall curve (AUCPR) compared with the decision tree on the original dataset and the Alghuried model [22].

Finally, Vangipuram et al. [30] proposed a model that overcomes the issue of missing values in the dataset (imputation). The model consists of two steps: imputation and classification. In the first step, the imputation algorithm groups the data into G1 and G2; G1 contains the instances with no missing values, while G2 consists of the instances with missing values. At this step, the marginal conditional probability and incremental clustering techniques have been used to produce an imputed dataset. In the second step, the classification algorithm used the obtained imputed dataset to classify the dataset into the desired class label. They used similar techniques as the imputation algorithm in this step, except G1 was used for training, and G2 was used for testing. The experiment compared two existing imputation techniques with eight classifiers, including the proposed one, applying them to the DS2OS traffic traces dataset [18]. The result demonstrated that the classification accuracy for the proposed model was in a range from 99 to 100%, which outperformed the existing models based on k-means imputation.

3.2 Unsupervised Learning

Unsupervised learning uses unlabelled data to find similarities between the data and set them into clusters or groups. Various unsupervised algorithms are employed to this end, for example hierarchical clustering (HCA) and PCA.

Zheng et al. [31] provided a solution to identify parking locations with high occupancy and possible faulty data. The proposed model was applied to parking data collected in San Francisco over a two-week period [32]. Different machine learning techniques were used: farthest first (FF) clustering, expectation maximization (EM) clustering, and one-class SVM: support vector data description (SVDD). SVDD was able to determine low and high occupancy, while FF was able to locate only heavy usage. EM helped to identify operational issues, such as sensor faults or parking spots affected by external factors.

For anomaly ranking and identification in sensor data relating to supercomputing, Morrow et al. [33] compared three algorithms: k-means, gaussian kernel density estimation, and DBSCAN. The chosen models utilised a Cray supercomputing dataset facility included in the Usenix Computer Failure Data Repository (CFDR) [34]. The outcomes revealed that DBSCAN was successful in avoiding false positives as well as anomaly identification identify. In addition, and based on location density, DBSCAN solely ranked genuinely anomalous data.

Garcia-Font et al. [35] compared four machine learning algorithms: mahalanobis distance (MD), local outlier factor (LOF), HCA, and one-class SVM (OC-SVM). Data was collected in the streets of Barcelona over a fourteen-day period of time and featured information state networks. Three metrics were used comparison of the algorithms involved three metrics: the TPR, FPR, and F-score. In addition, alternate feature selection procedures were utilised, e.g. Feature Vector 1 (FV1), Feature Vector 2 (FV2), and Feature

Vector 3 (FV3). FV1 included timestamp and sensor readings, FV2 contained FV1 plus the packet sequence number and battery power level. FV3 also contained FV2 and additional elements, such as the number of received MAC ACK and CTS. In cases which were more constrained (FPR < 5%), results revealed that OC-SVM with FV2 performed better, reaching over 75% in TPR.

As a consequence of sensors amassing huge volumes of unlabelled data, Martí et al. [36] put forward a model which combined both a segmentation algorithm (YASA) and OC-SVM. YASA is a time-series segmentation algorithm designed to overcome data issues, e.g. lack of consistency, noise and lack of reliability. The proposed model utilised YASA for homogenous data block identification which, subsequently, were utilised to feed the OC-SVM algorithm. 64 sensors provided data during the first six months of 2012. These were connected to an operational turbomachine. From a McNemar test, it was established that when combined, YASA and OC-SVM could outperform OC-SVM and statistical confidence intervals (CIs). Inoue et al. [37] proposed unsupervised learning deep neural networks (DNNs) and OC-SVM for the complex system. The models were applied to a Secure Water Treatment (SWaT) dataset [38, 39], with results showing that the DNN precision and F-scores were an improvement on those for an OC-SVM, though the OC-SVM was performed better in terms of recall.

Hoang and Nguyen [40] proposed a new distance formula for PCA to reduce computation overhead, and covered three issues: selection of principal components (PCs), distance measure, and quick anomaly detection. They applied a well-known distance, the Minkowski distance, with a threshold of 95%, by empirical cumulative distribution function (ECDF). The proposed model was applied to the Kyoto datasets 2006+ [41] with 14 features. When compared to other distance cases, the authors found that the model was acceptable in terms of TPR, FPR, and accuracy and produced similar results compared to other cases. However, the authors did not mention how the computation overhead is reduced in terms of processing time.

Finally, White and Legg [42] suggested a model for anomaly detection based on training only the normal instances through unsupervised learning using OC-SVM. They collected network data from IoT devices such as Amazon Echo and built a unique model for each device. The results found that the model could distinguish the normal and anomaly for different IoT devices regarding f-measure from 91.58 to 99.67%.

Table 1 summarizes the previously mentioned methods for anomaly detection in IoT. Most of the studies applied a single algorithm, either supervised or unsupervised, and other have suggested a combination of the two methods. In addition, different validation techniques were applied, such as accuracy, recall, or comparisons among algorithms. In supervised learning, available validation techniques are implemented, such as accuracy or FPR. Also, the proposed models can be compared to the most popular algorithms for detecting anomalies, such as SVMs, while unsupervised learning uses analysis and comparison to validate the proposed models.

However, supervised learning performs the task based on predefined labels annotated by human experts, which may be an infeasible or expensive task. It may require data to be labelled manually by human experts as a basis for decisions to predict the class labels as normal or abnormal for future data. Moreover, anomalies in the data are rare, which may take a long time to extract manually [3].

Although the performance of unsupervised algorithms in terms of accuracy is limited compared with supervised algorithms, the main advantage of unsupervised learning algorithms is the ability to deal with the absence of labelled data. However, the evaluation of unsupervised learning algorithms is still difficult, due to the lack of the ground truth. The

Table 1 Summary of supervised and unsupervised learning algorithms for anomaly detection in IoT

Learning technique	Machine learning algorithms	Results	Pros	Cons
Supervised-single and ensemble models	NNs [7]	93.75% in TPR	Easy to evaluate the results due to the availability of ground truth	It required labelled data which is labelled manually by the human
	RF, KNNs and J48 [14]	RF is better than KNNs and J48 in term of ROC curve		
	NNs, binary and multiclass SVMs [10]	SVMs is better than NNs in accuracy		
	NNs [9]	NNs is able to predicate with over 99%		
	NNs [13]	The accuracy is 99.4%		
	LR, DT, SVMs, RF, and NNs [17]	RF, DTs, and NNs are better than the other methods, achieving 99.4% accuracy		
Supervised-two-tier model	LDA and PCA, NB and KNNs [19]	Detection rate is 84.86%	Combining different algorithms may help to improve the performance of the detection	Models contain dimension reduction techniques such as PCA required ground truth
	IWC and C4.5 [22]	Accuracy is 97%		
	PCA and softmax regression [24]	Computing complexity is 5.72 seconds		
	DBSCAN and SVMs [26]	Accuracy detection is over 94%		
	K-means and SMO [27]	100% in accuracy and positive detection rate 0% in FPR		
	HAP and DT [28]	It is better in terms of FPR, recall, precision, and AUCPR compared with the DT and the Alghuried model [22]		
	Marginal conditional probability incremental clustering [30]	The accuracy is from 99 to 100%		

Table 1 (continued)

Learning technique	Machine learning algorithms	Results	Pros	Cons
Unsupervised	FF clustering EM clustering OC-SVM:(SVDD) [31]	EM is better to identify more operational issues	Deals with unlabelled data	Lack of the ground truth leads to complications in judging the quality of the clustering results and the performance is limited when compared with supervised algorithms
	YASA and OC-SVM [36]	Combination of YASA and OC-SVM is better than OC-SVM and CIs		
	K-means, gaussian kernel and DBSCAN [33]	DBSCAN is better than K-means and gaussian kernel		
	MD, LOF, HCA, OC-SVM [35]	OC-SVM is better with over 75% in TPR		
	DNNs OC-SVM [37]	DNNs is better in precision and F-score		
	A New PCA [40]	It is better in computation overhead		
	OC-SVM [42]	F-measure is from 91.58 to 99.67%		

most common unsupervised learning technique is clustering, which is used for exploratory data to find hidden patterns or groups. The quality of the cluster analysis can be measured by the following [43]:

- Compactness or cluster cohesion: the proximity to each other of data points in a cluster.
- Separation: how well-separated the clusters area, meaning how a cluster is a far away from other clusters.
- Connectivity or connectedness: which members are located in the same cluster as their nearest neighbours in the data space.

4 IoT Anomaly Detection Datasets

The data is the most important component of the machine learning process. Thus, in this section, we review the current IoT data repositories for anomaly detection. The previous studies reviewed in Sect. 3 deployed a variety of IoT datasets extracted from either networks or perceptual layers [44]. In the network layer, the data includes network information such as a communication network, satellite nets, and communication protocols. Whereas the perceptual layer contains the data collected from physical equipment, such as sensors or GPS. Some IoT datasets are publicly available, including the following:

- *Landsat satellite* [3]: contains intensity values for the images gained from satellite observations. The original task for this dataset is a multi-class task, while Golden delivers a new version of this dataset with the binary classification task for anomaly detection. The total of features is 36 in the range [0, 255], these features reflect four spectral bands where each band has nine values.
- *ISSDA* [8]: contains smart meter consumption data collected by the Irish Social Science Data Archive Centre between 2009 to 2011, with three attributes.
- *Citypulse project* [11, 12]: consists of pollution data collected from Aarhus, Denmark with 17,568 observations and eight attributes.
- *MIMIC* [15, 16]: contains physiological signals with a total of 12 attributes.
- *DS2OS* [18]: synthetic data was captured from IoT traffic traces by creating a virtual IoT environment consisting of normal and anomaly classes with 357,952 instances and 13 features. The anomaly class has eight types of attacks, including DOS attack.
- *Intel Lab* [23]: 2.3 million observations collected from 54 sensors with eight attributes, such as temperature, humidity, light, and voltage.
- *KDD Cup 99* [25]: consists of network information with 41 attributes that labels the data as normal or anomalous with 24 types of attacks.
- *NSL-KDD* [20]: a dataset that solves some of the problems found in KDD Cup 99 [25].
- *San Francisco parking* [32]: real data collected from sensors embedded in San Francisco parking from 13 August 2013 to 26 August 2013 with eight attributes.
- *LWSNDR* [29]: a real dataset collected from real sensors that are set up as indoor and outdoor sensors. It consists of four features, namely Reading number, Mote ID, temperature, and humidity.
- *CFDR* [34]: collects environmental data, which includes four attributes - ID number, timestamp, voltage, and temperature - using a supercomputer.
- *SWaT* [38, 39]: developed by all network information, sensor data, and actuator data with 36 types of attacks, but is considered a complicated system.

- *Kyoto 2006+* [41]: consists of network traffic from Kyoto University.

As shown in Table 2, labelled datasets are only available for network layers, except [18, 29]. However, a variety of perceptual-level datasets are available to the public, such as smart cities or healthcare, but most of these are not labelled. Moreover, some studies have collected data by simulating the sensors [9, 13, 35, 36, 42], yet none of them are published or available to the public. The information about the datasets is collected from research papers that have used these datasets or dataset repositories.

5 Future Research Directions

Machine learning algorithms (whether supervised or unsupervised) offer effective means of detecting, identifying, and classifying anomalies. The principal benefit of both approaches is the capability of learning from data that is freely and widely available in the environment of the IoT, though class labels (normal or anomalous) for IoT data are rarely available. Thus, it is a challenge to construct an anomaly detection model which is wholly reliant on supervised learning algorithms owing to a scarcity of labelled data. Additionally, the volume of IoT data is continually expanding at a rapid rate and the model must be able to predict anomalies for new and future data which, from the perspective of practicality, is restricted in unsupervised learning algorithms. Thus, in this section, we will put forward some potential research areas for future work.

Table 2 The current IoT data repositories for anomaly detection

Dataset	Labelled	Public	Type of the data		IoT application
			Network-based	Perceptual-based	Domain
Landsat satellite [3]	√	√	√		–
ISSDA [8]	×	√		√	Smart grid
Citypulse project [11, 12]	×	√		√	Smart city
MIMIC [15, 16]	×	√		√	Healthcare
DS2OS [18]	√	√		√	Different IoT domain such as smart home-smart phone
NSL-KDD [20]	√	√	√		–
Intel Lab [23]	×	√		√	Smart city
KDD Cup 99 [25]	√	√	√		–
San Francisco parking [32]	×	√		√	Smart city
CFDR [34]	×	√		√	Cyber-Physical System
LWSNDR [29]	√	√		√	Smart city
SWaT [38, 39]	√	√	√	√	Cyber-physical system
Kyoto 2006+ [41]	√	√	√		–

Cells marked (√), (×) and (−) mean available, unavailable or not mentioned, respectively

- *Providing labelled IoT data* Labelled data is the key element to validate and evaluate models. The previous studies reviewed in Sect. 4 presented a number of IoT datasets that do not contain normal and anomaly classes (unlabelled datasets), such as Intel Lab data [23], which lacks a ground truth. There are a few labelled IoT datasets for anomaly detection purposes, for example LWSNDR dataset [29]. To this end, the demand for labelled IoT datasets, either for a network or perceptual information with anomaly detection, needs to be addressed in future research.
- *Overcoming the issue of unlabelled data* The nurture of data in IoT is rapid fast and the data is formed as rows without classes. The unlabelled data in IoT represents a significant challenge, due to the cost of manual labelling, which requires time and experts. The proposed solutions in the literature [22, 26] are still not sufficient to overcome the issue of unlabelled data. While, [28] needs to enhance the classification process after automated the labelling process. More research should be carried out on this topic to overcome the issue of unlabelled data in IoT and reduce human intervention to label the data.
- *IoT benchmarking* IoT benchmarking is required. The term 'benchmarking' is used in machine learning to evaluate and compare different solutions based on different criteria, such as performance or computational time, with publicly available datasets. There are a few available comparative studies on IoT such as [35]; however, they do not provide or publish the data nor the algorithms to the public.

6 Conclusion

IoT data collection might suffer from anomalous behaviours, which are then translated into other issues, such as sensor faults or security issues. The detection of anomalies is widespread in several domains, for example fault or fraud detection. The main aim of detecting anomalies is the discovery of unusual behaviours or patterns that are distinct from the predicted behaviours. Machine learning algorithms are used to detect anomalies for the identification of normal and anomaly data classes.

This work has mainly focused on machine learning approaches for detection anomalies in the IoT and issues with the available IoT datasets. We have reviewed the existing machine learning algorithms (supervised or unsupervised) for IoT anomaly detection and discussed the main challenges of both approaches. Finally, new future research directions have been proposed and discussed from the perspective of the dataset.

Author Contributions NA carried out the conceptualization, formal analysis, methodology, implementation, visualization, and writing the original draft of the manuscript. RA carried out the conceptualization, formal analysis, methodology, revising and editing the paper. SB revised and edited the paper.

Data Availability Data sharing not applicable to this article as no datasets were generated or analysed during the current study

Declaration

Conflict of interest The authors declare that they have no conflcit of interests.

References

1. Cisco global cloud index: Forecast and methodology, 2016–2021 white paper (2018). https://www.cisco.com/c/en/us/solutions/collateral/service-provider/global-cloud-index-gci/white-paper-c11-738085.html
2. Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys, 41*(3), 15:1. https://doi.org/10.1145/1541880.1541882.
3. Goldstein, M., & Uchida, S. (2016). A comparative evaluation of unsupervised anomaly detection algorithms for multivariate data. *PLoS ONE, 11*(4), 1. https://doi.org/10.1371/journal.pone.0152173.
4. Ahmed, M., Mahmood, A. N., & Hu, J. (2016). A survey of network anomaly detection techniques. *Journal of Network and Computer Applications, 60,* 19.
5. Tsai, C. F., Hsu, Y. F., Lin, C. Y., & Lin, W. Y. (2009). Intrusion detection by machine learning: A review. *Expert Systems with Applications, 36*(10), 11994. https://doi.org/10.1016/j.eswa.2009.05.029.
6. Jang, J. S. R., Sun, C. T., & Mizutani, E. (1997). *Neuro-fuzzy and soft computing: A computational approach to learning and machine intelligence*. New Jersey: Prentice Hall.
7. Ford, V., Siraj, A., & Eberle, W. (2014). In *2014 IEEE symposium on computational intelligence applications in smart grid (CIASG)* (pp. 1–6). https://doi.org/10.1109/CIASG.2014.7011557
8. Commission for Energy Regulation (CER), Irish Social Science Data Archive (ISSDA). (2012). www.ucd.ie/issda/data/commissionforenergyregulationcer/
9. Cañedo, J., & Skjellum, A. (2016). In *2016 14th Annual conference on privacy, security and trust (PST)* (pp. 219–222). https://doi.org/10.1109/PST.2016.7906930
10. Jain, R., & Shah, H. (2016). In *2016 International conference on signal and information processing (IConSIP)* (pp. 1–5). https://doi.org/10.1109/ICONSIP.2016.7857445
11. Ali, M. I., Gao, F., & Mileo, A. (2015). In *Proceedings of ISWC 2015–14th international semantic web conference (W3C* (pp. 374–389). Bethlehem, PA
12. Pollution Data, Citypulse Project. (2014). http://iot.ee.surrey.ac.uk:8080/datasets.html
13. Hodo, E., Bellekens, X., Hamilton, A., Dubouilh, P., Iorkyase, E., Tachtatzis, C., & Atkinson, R. (2016). In *International symposium on networks. Computers and communications (ISNCC)* (Vol. 2016, pp. 1–6). https://doi.org/10.1109/ISNCC.2016.7746067
14. Pachauri, G., & Sharma, S. (2015) Procedia Computer Science *70*, 325. https://doi.org/10.1016/j.procs.2015.10.026. (Proceedings of the 4th International Conference on Eco-friendly Computing and Communication Systems).
15. Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., et al. (2000). Circulation electronic pages. *Circulation, 101*(23), e215. https://doi.org/10.1161/01.CIR.101.23.e215.
16. PhysioNet. https://www.physionet.org/cgi-bin/atm/ATM
17. Hasan, M., Islam, M. M., Zarif, M. I. I., & Hashem, M. (2019). *Internet of Things, 7,* 100059.
18. Pahl, M. O., & Aubet, F. X. (2018). In *2018 14th International conference on network and service management (CNSM)* (pp. 72–80).
19. Pajouh, H. H., Javidan, R., Khayami, R., Ali, D., & Choo, K. K. R. (2016). A two-layer dimension reduction and two-tier classification model for anomaly-based intrusion detection in IoT backbone networks. *IEEE Transactions on Emerging Topics in Computing*. https://doi.org/10.1109/TETC.2016.2633228.
20. Tavallaee, M., Bagheri, E., Lu, W., & Ghorbani, A. A. (2009). In *2009 IEEE symposium on computational intelligence for security and defense applications* (pp. 1–6). https://doi.org/10.1109/CISDA.2009.5356528
21. Pajouh, H. H., Dastghaibyfard, G., & Hashemi, S. (2017). Two-tier network anomaly detection model: A machine learning approach. *Journal of Intelligent Information Systems, 48*(1), 61. https://doi.org/10.1007/s10844-015-0388-x.
22. Alghuried, A. (2017). A model for anomalies detection in internet of things (IOT) using inverse weight clustering and decision tree. Masters dissertation. https://doi.org/10.21427/D7WK7S
23. Bodik, P., Hong, W., Guestrin, C., Madden, S., Paskin, M., & Thibaux, R. (2004). Intel Lab Data. http://db.csail.mit.edu/labdata/labdata.html
24. Zhao, S., Li, W., Zia, T., & Zomaya, A. Y. (2017). In *2017 IEEE 15th international conference on dependable, autonomic and secure computing, 15th international conference on pervasive intelligence and computing, 3rd international conference on big data intelligence and computing and cyber science and technology congress(DASC/PiCom/DataCom/CyberSciTech)* (pp. 836–843). https://doi.org/10.1109/DASC-PICom-DataCom-CyberSciTec.2017.141

25. KDD Cup 1999 Data. (1999). http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html
26. SaeediEmadi, H., & Mazinani, S. M. (2018). A novel anomaly detection algorithm using DBSCAN and SVM in wireless sensor networks. *Wireless Personal Communications, 98*(2), 2025. https://doi.org/10.1007/s11277-017-4961-1.
27. Hosseini, M., & Borojeni, H. R. S. (2018). In *Proceedings of the international conference on smart cities and internet of things* (SCIOT 18). Association for Computing Machinery. https://doi.org/10.1145/3269961.3269975
28. Alghanmi, N., Alotaibi, R., & Buhari, S. M. (2019). HLMCC: A hybrid learning anomaly detection model for unlabeled data in internet of things. *IEEE Access, 7,* 179492. https://doi.org/10.1109/ACCESS.2019.2959739.
29. Suthaharan, S., Alzahrani, M., Rajasegarar, S., Leckie, C., & Palaniswami, M. (2010). In *2010 Sixth international conference on intelligent sensors, sensor networks and information processing* (pp. 269–274). https://doi.org/10.1109/ISSNIP.2010.5706782
30. Vangipuram, R., Gunupudi, R. K., Puligadda, V. K., & Vinjamuri, J. (2020). A machine learning approach for imputation and anomaly detection in IoT environment. *Expert Systems, 37*(5), e12556. https://doi.org/10.1111/exsy.12556.
31. Zheng, Y., Rajasegarar, S., Leckie, C., & Palaniswami, M. (2014). In *2014 IEEE ninth international conference on intelligent sensors, sensor networks and information processing (ISSNIP)* (pp. 1–6). https://doi.org/10.1109/ISSNIP.2014.6827618
32. San Francisco Parking Data. (2013). http://sfpark.org/
33. Morrow, A., Baseman, E., & Blanchard, S. (2016). In *2016 International conference on computational science and computational intelligence (CSCI)* (pp. 629–632). https://doi.org/10.1109/CSCI.2016.0124
34. Schroeder, B., & Gibson, G. A. (2007). *Workshop on reliability analysis of system failure data (RAF07)*. Cambridge: MSR Cambridge.
35. Garcia-Font, V., Garrigues, C., & Rifà-Pous, H. (2016). A comparative study of anomaly detection techniques for smart city wireless sensor networks. *Sensors, 16,* 6. https://doi.org/10.3390/s16060868.
36. Martí, L., Sanchez-Pi, N., Molina, J. M., & Garcia, A. C. B. (2015). Anomaly detection based on sensor data in petroleum industry applications. *Sensors, 15*(2), 2774. https://doi.org/10.3390/s150202774.
37. Inoue, J., Yamagata, Y., Chen, Y., Poskitt, C. M., & Sun, J. (2017). In *2017 IEEE international conference on data mining workshops (ICDMW)* (pp. 1058–1065). https://doi.org/10.1109/ICDMW.2017.149
38. Secure Water Treatment (SWaT). (2017). http://itrust.sutd.edu.sg/testbeds/secure-water-treatment-swat/
39. Goh, J., Adepu, S., Junejo, K. N., & Mathur, A. (2017). *Critical information infrastructures security* (pp. 88–99). Cham: Springer.
40. Hoang, D. H., & Nguyen, H. D. (2018). In *2018 20th International conference on advanced communication technology (ICACT)* (pp. 381–386). https://doi.org/10.23919/ICACT.2018.8323766
41. Traffic Data from Kyoto University's Honeypots. (2006). http://www.takakura.com/Kyoto_data
42. White, J., & Legg, P. (2021). In *2021 International conference on cyber situational awareness, data analytics and assessment (CyberSA)* (pp. 1–8). https://doi.org/10.1109/CyberSA52016.2021.9478248
43. Handl, J., Knowles, J., & Kell, D. B. (2005). Computational cluster validation in post-genomic data analysis. *Bioinformatics, 21*(15), 3201.
44. Suo, H., Wan, J., Zou, C., & Liu, J. (2012). In *2012 International conference on computer science and electronics engineering* (Vol. 3, pp. 648–651). https://doi.org/10.1109/ICCSEE.2012.373

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Nusaybah Alghanmi Nusaybah Alghanmi received her B.S. and M.S. in Information Technology from King Abdulaziz University in Jeddah, Saudi Arabia. She is a teaching assistant at Faculty of Computing and Information Technology, University of Jeddah, Khulais, Saudi Arabia. Her research interests include Wireless Sensor Networks, Machine Learning and Data Mining.

Reem Alotaibi Reem Alotaibi is an associate professor at the Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia. Currently, she is the supervisor of the Information Technology department. Dr. Alotaibi received her Ph.D. in computer science from University of Bristol, Bristol, U.K., in 2017. During 2017–2018 she was a visiting lecturer at the Intelligent Systems Laboratory, University of Bristol. Her research interests include Artificial Intelligence, Machine learning and Data mining, Statistical analysis.



Seyed M. Buhari Seyed M. Buhari is an associate professor, Department of Information Technology, Faculty of Computing and Information Technology, King Abdulaziz University. His area of interest is Data Mining, Grid Computing and Wireless sensor Networks.